

SYNTHETIC IDENTITY FRAUD DETECTION

Digital Lending Onboarding via KYC Verification & Behavioral Signals

Final Year Capstone Project — BFSI / AI-ML Domain
Machine Learning Pipeline & Live Interactive Web Dashboard

Executive Summary & Problem Statement

The Threat: Synthetic Identity Fraud

- Combines real PII (valid PAN) with fake details (burner phone, fake address).
- Fastest-growing financial crime category in digital lending onboarding.
- Bypasses traditional static KYC checks because individual field queries return clean responses.
- No individual victim exists initially to raise an alert (credit bust-out risk).

Our Machine Learning Solution

- **Multimodal Risk Fusion:** Blends KYC consistency, identity freshness, behavioral biometrics, and device velocity.
- **Machine Learning Engine:** Random Forest Classifier tuned with balanced class weights.
- **Real-Time Scoring:** Live risk tiering (<30% Low, 30-60% Medium, >60% High).
- **Streamlit Web App:** Interactive onboarding decision interface.

Feature Engineering & 12 Core Signals

1. KYC Consistency

- name_address_mismatch_score
- dob_pan_mismatch
- document_reuse_count

3. Behavioral Telemetry

- session_fill_time_sec
- typing_speed_variance

2. Identity Freshness

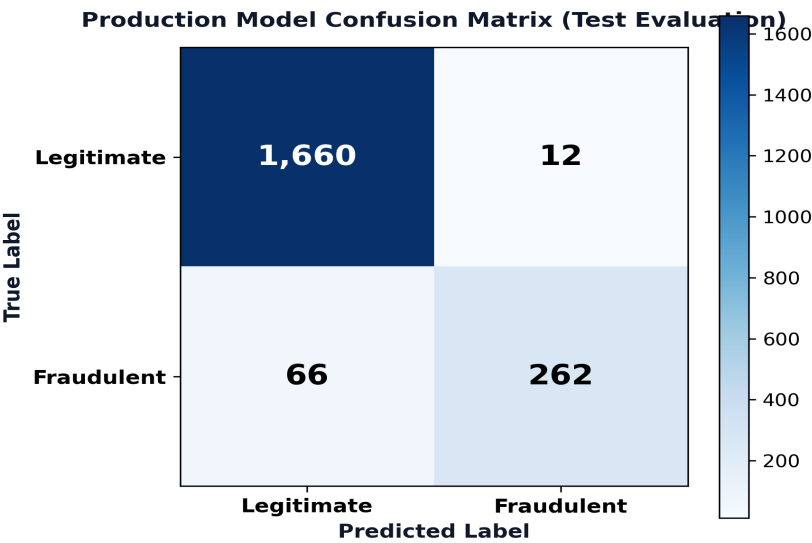
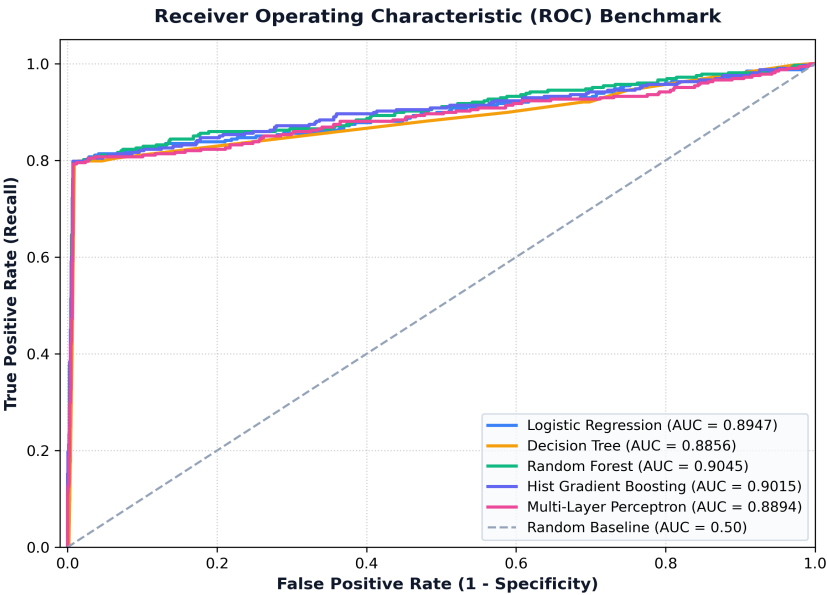
- phone_age_days
- email_age_days
- credit_bureau_hit
- social_footprint_score

4. Device & Network Velocity

- device_reuse_across_apps
- application_velocity_24h
- ip_geolocation_mismatch

Model Evaluation Results & Key Metrics

ROC-AUC: 0.8834 | Fraud Precision: 0.9500 | Fraud Recall: 0.7500 | Fraud F1: 0.8400



Top Predictive Signals (Feature Importances)

Key Feature Insights:

- 1. **Email Age (0.232):** Freshly registered emails (<10 days) heavily correlate with fraud.
- 2. **Phone Number Age (0.220):** Burner lines purchased recently indicate identity creation.
- 3. **Name/Address Mismatch (0.182):** Distance score catches spliced identity credentials.
- 4. **Social Footprint (0.097):** Zero digital presence reflects synthetic profiles.
- 5. **Session Fill Time (0.096):** Scripted/rushed form completion indicates automated bots.

